

THIRD SEMESTER										
Course Code	Title of the Course	Exam (Hrs)	Hours				Max. Marks			Credits
			L	T	P	Total	CA	CE	Total	
25CAP317	Major Course –Artificial Intelligence	3	4	0	0	4	25	75	100	4
25CAP318	Major Course –Machine Learning	3	4	0	0	4	25	75	100	4
25CAP319	Major Course – Full Stack Development	-	3	1	0	4	100	-	100	4
25CAP320A/ 25CAP320B	Major Elective Course II Software Quality Assurance / Agile Methodologies	3	4	0	0	4	25	75	100	4
25CAP321	Major Course –Practical-VII (Artificial Intelligence Lab)	3	0	0	3	3	40	60	100	2
25CAP322	Major Course –Practical-VIII (Machine Learning Lab) [Skill based]	3	0	0	2	2	40	60	100	2
25CAP323	Major Course –Practical-IX (Full Stack Development Lab) [Skill based]	3	0	0	4	4	40	60	100	4
25CAP324	Major Course – Project I Viva Voce	-	-	-	-	-	40	60	100	4
25CAP325	Inter Disciplinary Course - Theory	3	3	0	0	3	25	75	100	2
25CAP326	Inter Disciplinary Course – Practical [Skill based]	3	0	0	2	2	40	60	100	2
TOTAL			18	1	11	30	400	600	1000	32

FOURTH SEMESTER										
Course Code	Title of the Course	Exam (Hrs)	Hours				Max. Marks			Credits
			L	T	P	Total	CA	CE	Total	
25CAP427	Major Course - Project II / Innovation / Start-Up	-	-	-	-	-	80	120	200	6
TOTAL			-	-	-	-	80	120	200	6

Non CGPA Credit Course (NCCC) Any one Massive Open Online Course (MOOC)*	1	2	I to III
	CA	CE	Total
Grand Total	1145	1755	2900
			Credits
			96

***Massive Open Online Course (MOOC)** should be completed during I to III semester. Students shall complete any **one MOOC** available on SWAYAM-NPTEL / Coursera / any other MOOC portal. The course must be of minimum 4 weeks / 20 hours duration. The Online Course to be completed at PG level should be different from the one completed during the UG Programme.

Skill Enhancement & Employability Development Programme (SEED) offered by the College for a period of **80 hours (out of class hours)** during I (26h), II (26h) and III (28h) semesters is mandatory to qualify for the award of Degree.

Compulsory Component to be completed by the students	To be completed during
Soft Skill Training	II
One Certificate Course offered by Computer Stream Programmes	II - III

Inter Disciplinary Course (IDC) offered by the Department of Computer Applications (PG)

Course Code	Title of the Course
25IDCCAPT	Inter Disciplinary Course – Data Analytics (theory)
25IDCCAPP	Inter Disciplinary Course – (Data Analytics Lab) [Practical]

List of Certificate Courses (as Value Added Courses)

<u>S.No</u>	Course Title	Offered by
1	Web Designing	Department of Computer Science
2	Basics of Japanese Language & Web Development in Java	
3	AI Automation Tools	Department of Computer Science with Data Analytics
4	Software Testing and Automation Tools	Department of Information Technology
5	UI Design and Development	Department of Networking and Mobile Application
6	Full Stack Development for Beginners	Department of Software Systems
7	Graphic Design & Multimedia	Department of Computer Applications-BCA
8	Foundations of Data Science	Department of Computer Applications-MCA
9	Data Visualization in Business Intelligence	Department of Computer Technology
10	Digital Tools for Documentation	Department of Artificial Intelligence and Machine Learning

Major Courses offered to other departments	
Programme	M.Com (Commerce with Computer Applications)
Course Code	Course Name
25CCP317	Major Course - RDBMS
25CCP318	Major Course Computer Practical -III RDBMS [Skill Based]
25CCP427	Major Course – Python Programming
25CCP428	Major Course Computer Practical -V Python Programming [Skill Based]

Inter Disciplinary Course – Cluster Details

<u>S.No</u>	Course Title	Offered by
1		Mathematics (Unaided)
2		Applied Electronics
3		MSc Software Systems
4		MCom International Business
5		MCom Corporate Secretaryship
6		MCom Computer Applications
7	Inter Disciplinary Course – Data Analytics Inter Disciplinary Course – (Data Analytics Lab) [Skill based]	MCA

Course Code	25CAP317	L	T	P	C
Course Title	Artificial Intelligence	4	-	-	4
Major Course		Employability, Entrepreneurship & Skill Development			
Pre-requisite	Problem solving & Programming, Basics of Machine Learning	Regulation 2025-2026			
Class	II MCA	Semester	III		
Course Outcomes	On completion of this course, students will be able to			K Level	
	CO1: identify appropriate AI methods to solve a given problem.			K1, K2, K3	
	CO2: formalize a given problem in the framework of different AI methods.			K1, K2, K3, K4	
	CO3: find out solutions for undesirable end.			K1, K2, K3, K4, K5	
	CO4: get an insight in designing prototype for real life problems.			K1, K2, K3, K4, K5	
	CO5: develop an insight in development a simple robot model.			K1, K2, K3, K4, K5, K6	

SYLLABUS

Module	Contents	No. of Hours
I	Foundations of AI: History of AI. Intelligent Agents: Agents and environments - The nature of Environments. Problem Solving: Problem solving Agents – Example Problems – Searching for Solutions – Uninformed Search Strategies – Informed Search Strategies – Heuristic Functions.	10
II	Adversarial search: Games – Optimal Decisions in Games – Alpha – Beta Pruning – Imperfect Real – Time Decisions – Stochastic Games. Knowledge reasoning and planning: Logical Agents: Knowledge-based Agents – The Wumpus World – Logic- Propositional Logic: A Very Simple Logic. First order Logic: Representation Revisited Syntax and Semantics of First-Order Logic - Using First-order logic. Inference in First -Order Logic: Propositional vs. First-Order Inference – Unification and Lifting - Forward Chaining - Backward Chaining.	12
III	Learning: Learning from Examples - Forms of Learning – Supervised Learning - Learning Decision Trees – Artificial Neural Networks.	8
IV	Communicating, perceiving and acting: Natural Language Processing: Language Models - Text Classification - Information Retrieval - Information Extraction. Natural Language for Communication: Phrase Structure Grammars - Syntactic Analysis - Augmented Grammars and Semantic Interpretation - Machine Translation - Speech Recognition. Perception: Image Formation - Early Image Processing Operations.	9
V	Robotics: Introduction - Robot Hardware - Robotics Perception - Planning to Move - Planning Uncertain Movements - Moving. Case Study on Robotic Software Architectures, Case studies of Artificial Intelligence in Indian Knowledge Systems such as strategic games, and language processing.	9
	Total Hours	48

References		Essential: [Text Book] 1. Stuart Russell and Peter Norvig, “Artificial Intelligence: A Modern Approach”, Third Edition, Pearson Education, Reprint 2025. [Module I-V] Suggestive: 1. Nilsson, N.J., “Artificial Intelligence and New Systems”, First Edition, Elsevier, 2022. 2. Patterson, D.W., Introduction to Artificial Intelligence and Expert Systems, Prentice Hall of India, 2015. 3. Denis Rothman, “Artificial Intelligence by Example: Acquire advanced AI, machine learning, and deep learning design skills”, Second Edition, 2020. 4. Anuradha Srinivasa Raghavan, Vincy Joseph, “Machine Learning”, 2019. 5. Deepak Khemani, “A First Course in Artificial Intelligence”, 2017. Online Resources: 1. https://www.tutorialspoint.com/artificial_intelligence/index.htm 2. https://www.geeksforgeeks.org/artificial-intelligence/what-is-an-ai-technique/			
Teaching-Learning Process		1.Chalk and Talk 2.Power Point Presentation 3. Audio, Video Tools 4. Seminars 5. Quiz			
Assessment Methods		1. Problem based assignments 2. Practical assignments laboratory reports 3. Observation of practical skills 4. Team project reports 5. Oral presentation including seminar presentation			
Designed by	Dr.S. Manju	Verified by	Dr.V.S. Anita Sofia	HoD	Dr.L. Thara

Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12		PSO1	PSO2	PSO3
CO1	2	3	3	3	3	-	-	3	3	1	2	-		1	3	-
CO2	3	3	3	3	3	2	-	-	2	-	3	3		2	3	2
CO3	3	3	3	3	2	2	1	-	-	-	3	3		3	3	-
CO4	3	-	2	2	3	-	-	2	3	-	3	-		3	3	3
CO5	-	3	-	2	-	1	-	3	-	-	3	-		3	-	-

* 1 = Low; 2 = Moderate; 3 = Strong

Course Code	25CAP318	L	T	P	C
Course Title	Machine Learning	4	-	-	4
Major Course		Employability, Entrepreneurship & Skill Development			
Pre-requisite	Problem Solving and Programming	Regulation 2025-2026			
Class	II MCA	Semester	III		
Course Outcomes	On completion of this course, students will be able to:		K level		
	CO1: explain the fundamental concepts of machine learning, learning paradigms, data preprocessing methods, and the ML workflow.		K1, K2, K3		
	CO2: apply supervised learning algorithms such as regression, classification, SVM, and decision trees to solve real-world prediction problems.		K1, K2, K3, K4		
	CO3: analyze datasets using unsupervised learning techniques, including clustering, association rule mining, and dimensionality reduction.		K1, K2, K3, K4, K5		
	CO4: evaluate and optimize machine learning models using appropriate performance metrics, validation techniques, and hyperparameter tuning methods.		K1, K2, K3, K4, K5		
	CO5: develop and implement machine learning solutions using Python tools and libraries for practical applications across domains.		K1, K2, K3, K4, K5, K6		

SYLLABUS

Module	Contents	No. of Hours
I	Introduction to Machine Learning & Data Preprocessing — Definition, Scope, Applications - Types of Learning — Supervised, Unsupervised, Reinforcement - ML Pipeline and Workflow - Data Collection and Data Cleaning - Data Preprocessing Techniques - Feature Selection and Feature Engineering - Train-Test Split and Cross Validation -Overfitting and Underfitting - Bias-Variance Trade-off.	9
II	Supervised Learning Algorithms - Regression Concepts - Linear Regression - Simple & Multiple - Logistic Regression - k-Nearest Neighbors (KNN) - Decision Trees - Naïve Bayes Classifier - Support Vector Machines (SVM) - Model Evaluation Metrics - Accuracy, Precision, Recall, F1 Score - Confusion Matrix - ROC Curve & AUC.	10
III	Unsupervised Learning & Clustering - Unsupervised Learning Overview - Clustering Concepts - K-Means Clustering - Hierarchical Clustering - Density-Based Clustering (DBSCAN) - Association Rule Mining - Apriori Algorithm - Dimensionality Reduction - PCA (Principal Component Analysis) -Feature Extraction Techniques.	10
IV	Advanced Machine Learning Methods - Ensemble Learning – Bagging - Boosting - Random Forest - Gradient Boosting - Model Optimization - Hyperparameter Tuning - Grid Search & Random Search - Neural Networks Basics – Perceptron - Multilayer Perceptron - Introduction to Deep Learning - Introduction to Reinforcement Learning Concepts.	10
V	Practical ML, Tools & Applications - ML with Python - NumPy, Pandas - Matplotlib & Seaborn - ML using Scikit-learn - Model Deployment Basics - ML in Real-World Applications – Healthcare, Finance, IoT & Smart Systems. Case studies and applications of ML in Indian Knowledge Systems such as traditional medicine, agriculture, language processing, and astronomy.	9
	Total Hours	48

References	Essential: [Text Book] 1. Ruye Wang, “Introduction to Machine Learning: From Math to Code”, Cambridge University Press (Dec 2025) (Modules: I & II) 2. Wenmin Wang, “Principles of Machine Learning: The Three Perspectives”, Springer Nature 2024 (Modules: II & III) 3. Vijay Janapa Reddi, “Machine Learning Systems: Principles and Practices of Engineering AI Systems”, MIT Press, 2026 (Modules: III to V) Suggestive: 1. Ethem Alpaydin, "Introduction to Machine Learning", MIT Press, Prentice Hall of India, Third Edition 2014. 2. Richard S. Sutton and Andrew G. Barto,” Reinforcement Learning: An Introduction (Adaptive Computation and Machine Learning series)” 2nd edition, 2018. 3. Mehryar Mohri, Afshin Rostamizadeh, Ameet Talwalkar, "Foundations of Machine Learning”, MIT Press, 2012. 4. Charu C. Aggarwal, “Data Classification Algorithms and Applications”, CRC Press, 2014 Online Resources: 1. https://www.coursera.org/specializations/machine-learning-introduction 2. https://www.simplilearn.com/skills-required-to-become-a-full-stack-developer-article 3. https://www.w3schools.com/whatis/whatis_fullstack.asp#:~:text=A%20full%20stack%20web%20developer,ASP%2C%20Python%2C%20or%20Node				
Teaching-Learning Process	1. Chalk and talk 2. PowerPoint presentations 3. Projects 4. Audio & video tools 5. Quiz				
Assessment Methods	1. Oral presentation including seminar presentation 2. Problem based assignments 3. Open-book tests 4. Case-study reports 5. Team project reports				
Designed by	Dr.V.S. Anita Sofia	Verified by	Dr.S. Manju	HoD	Dr.L. Thara

Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12		PSO1	PSO2	PSO3
CO1	2	3	3	3	3	-	-	3	3	-	2	-		1	3	-
CO2	3	3	3	3	3	2	-	-	2	-	3	3		2	3	2
CO3	3	3	3	3	3	2	1	-	-	-	3	3		3	3	-
CO4	3	-	2	2	3	-	-	2	3	-	3	-		3	3	3
CO5	-	3	-	2	-	1	-	3	-	-	3	-		3	-	-

* 1 = Low; 2 = Moderate; 3 = Strong

Course Code	25CAP319	L	T	P	C
Course Title	Full Stack Development	3	1	-	4
Major Course		Employability, Entrepreneurship & Skill Development			
Pre-requisite	Knowledge of C, OOPS, Java, Python, Web Framework	Regulation 2025-2026			
Class	II MCA	Semester	III		
Course Outcomes	On completion of this course, student will be able to:		K Level		
	CO1: explain and apply the fundamentals of web programming and client-side scripting.		K1, K2, K3		
	CO2: demonstrate and analyze server-side development using Node.js.		K1, K2, K3, K4		
	CO3: develop and evaluate APIs using the Express framework.		K1, K2, K3, K4, K5		
	CO4: design and implement database solutions using SQL and NoSQL databases.		K1, K2, K3, K4, K5		
	CO5: design and build dynamic web applications using advanced client-side scripting and the ReactJS framework.		K1, K2, K3, K4, K5, K6		

SYLLABUS

Module	Contents	No. of Hours
I	Introduction to CSS and JavaScript Introduction to Web: Server - Client - Communication Protocol (HTTP) – Structure of HTML Documents – Basic Markup tags – Working with Text and Images with CSS– CSS Selectors – CSS Flex box - JavaScript: Data Types and Variables - Functions - Events – AJAX: GET and POST.	9
II	Server-Side Programming with Node JS Introduction to Web Servers – JavaScript in the Desktop with NodeJS – NPM – Serving files with the http module – Introduction to the Express framework – Server-side rendering with Templating Engines – Static Files - async/await - Fetching JSON from Express.	10
III	Advanced Node JS and Database Introduction to No SQL databases – MongoDB system overview - Basic querying with MongoDB shell – Request body parsing in Express – Node JS MongoDB connection – Adding and retrieving data to MongoDB from Node JS – Handling SQL databases from Node JS – Handling Cookies in Node JS – Handling User Authentication with Node JS.	10
IV	Advanced Client-Side Programming React JS: React DOM - JSX - Components - Properties – Fetch API - State and Lifecycle -JS Local storage - Events - Lifting State Up - Composition and Inheritance.	10
V	APP Implementation in Cloud Cloud providers Overview – Virtual Private Cloud – Scaling (Horizontal and Vertical) – Virtual Machines, Ethernet and Switches – Docker Container – Kubernetes.	9
	Total Hours	48

References	Essential : [Text Books] 1. David Flanagan, "Java Script: The Definitive Guide", O'Reilly Media, Inc, 7th Edition, 2020, [Modules: I & II] 2. Matt Frisbie, "Professional JavaScript for Web Developers", Wiley Publishing, Inc, 4th Edition, ISBN: 978-1-119-36656-0, 2019, [Modules: II to V] Suggestive: 1. Alex Banks, Eve Porcello, "Learning React", O'Reilly Media, Inc, 2nd Edition, 2020. 2. Marc Wandschneider, "Learning Node", Addison-Wesley Professional, 2nd Edition, 2016. 3. Joe Beda, Kelsey Hightower, Brendan Burns, "Kubernetes: Up and Running", O'Reilly Media, 1st edition, 2017. 4. Paul Zikopoulos, Christopher Bienko, Chris Backer, Chris Konarski, Sai Vennam, "Cloud without Compromise", O'Reilly Media, 1st edition, 2021. Online Resources: 1. https://www.coursera.org/learn/backend-development-nodejs-express-mongodb-restapis 2. https://www.simplilearn.com/skills-required-to-become-a-full-stack-developer-article2 3. https://www.w3schools.com/whatis/whatis_fullstack.asp#:~:text=A%20full%20stack%20web%20developer,ASP%2C%20Python%2C%20or%20Node				
Teaching-Learning Process	1. Chalk and talk 2. PowerPoint presentations 3. Projects 4. Audio and video tools 5. Quiz				
Assessment Methods	1. Seminar 2. Library Work 3. Team Project 4. Case Study Reports 5. Problem-based Assignments				
Designed by	Dr. V. S. Anita Sofia	Verified by	Dr.S. Manju	HoD	Dr.L. Thara

Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12		PSO1	PSO2	PSO3
CO1	3	3	3	3	3	-	-	3	3	-	2	-		1	3	-
CO2	3	3	3	3	3	3	-	-	2	-	3	3		2	3	2
CO3	3	3	3	3	3	2	1	-		-	3	3		3	3	-
CO4	3	-	2	2	3	-	-	2	3	-	3	-		3	3	3
CO5	-	3	-	2	-	1	-	3	-	-	3	-		3	-	-

* 1 = Low; 2 = Moderate; 3 = Strong

Course Code	25CAP320A	L	T	P	C
Course Title	Software Quality Assurance	4	-	-	4
Major Elective Course II		Employability & Skill Development			
Pre-requisite	Introductory concepts in testing methodologies and project management	Regulation 2025-2026			
Class	II MCA	Semester	III		
Course Outcomes	On completion of this course, student will be able to:			K Level	
	CO1: recognize fundamental concepts and philosophies of quality and quality management.			K1, K2, K3	
	CO2: analyze quality standards, models, frameworks and CMMI.			K1, K2, K3	
	CO3: apply quality assurance, verification, validation, and testing practices in Software projects.			K1, K2, K3, K4	
	CO4: evaluate software quality using qualitative and quantitative tools, metrics and continuous improvement techniques.			K1, K2, K3, K4, K5	
	CO5: evaluate software project and product risks by applying appropriate risk assessment and mitigation strategies.			K1, K2, K3, K4, K5, K6	

SYLLABUS

Module	Contents	No. of Hours
I	Fundamentals of Software Quality: Introduction to concepts of quality – Historical perspective of quality improvements – Meaning and definitions of quality – Customer expectations and what customers want from a product – Quality views from different perspectives and stakeholders – Core components of product quality – Customer, supplier, management and employee views – Quality gaps – Total Quality Management (TQM) concepts. Quality Improvement Approaches: ‘q’ and ‘Q’ organizations – Shewhart’s PDCA cycle – Deming’s approach to quality improvement – Six Sigma quality – Quality control, quality assurance and quality management – Measurement and statistical analysis – Cultural change for quality improvement.	10
II	Quality Standards and Models: Types of Standards – Writing and Documentation of Standards – Quality Models and Standards – Difference Between Model and Standard – Standards and Guidelines. International and National Quality Models: ISO Standards - Capability Maturity Model Integration (CMMI) – Malcolm Baldrige National Quality Award – Organizational Inputs and Quality Challenges.	10
III	Quality Management at Organizational Level: Software Characteristics – Software/System Development Lifecycle – Product Classification – Problematic Areas of SDLC – Process Groups – Project-Related Planning Guidelines. Quality Assurance and Planning: Quality Assurance Concepts – Quality Control vs Quality Assurance – Role of Quality Assurance Analyst – Quality Function Deployment – Quality Plans – Deployment of Quality Function – Cultural Change Management.	10

IV	Software Verification, Validation and Testing: Verification Methods – Reviews, Walkthroughs, Inspections and Audits – Validation Concepts – Levels of Testing – Acceptance Testing – Test Planning – Defect Management – Test Reports. Qualitative and Quantitative Analysis: Measurement Systems – Types of Data – Quality Tools – Balanced Scorecard – Benchmarking – Quality Function Deployment – Metrics and Continuous Improvement.				10
V	Risk Analysis and Risk Management: Introduction to risk – Project risks – Product risks – Constraints– Risk analysis process – Types of software risks – Risk handling and mitigation strategies – Compliance aspects and Sarbanes – Enterprise Risk Management (ERM) overview. Auditing and Control Mechanisms: Audit concepts – Internal audit – Control and internal system control – Levels of controls – McFarlan’s strategic grid of control management – Application controls – Accounting standards – Building controls into software systems – COSO model – Characteristics of effective control – COBIT – ITIL.				8
	Total Hours				48
References	Essential: [Text Book] 1. Milind Limaye, “Software Quality Assurance”, McGraw Hill Higher Education, Reprint 2020. Suggestive: 1. Dr. Dinesh Jain, Dr. Abhay Kothari, “Software Testing and Quality Assurance”, BPB Publications, 2025. 2. Grace Mary R. Sharmila, “Software Quality Management”, First Edition, Sams Publisher, 2012. Online Resources: 1. https://www.w3schools.com 2. https://www.iso.org/standards.html 3. https://www.isaca.org/				
Teaching-Learning Process	1. Chalk and talk 2. PowerPoint presentations 3. Projects 4. Audio and video tools 5. Quiz				
Assessment Methods	1. Problem based assignments 2. Observation of Practical skills 3. Practical Assignment Laboratory reports 4. Individual Project reports 5. Problem based Assignments				
Designed by	Dr. Thiagarajan	Verified by	Dr.L. Thara	HoD	Dr.L. Thara

Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12		PSO1	PSO2	PSO3
CO1	3	2	2	1	2	-	-	-	2	2	-	2		3	2	-
CO2	3	3	3	2	3	-	2	-	3	2	2	2		3	3	-
CO3	3	2	3	2	2	-	-	2	3	2	1	2		3	2	-
CO4	2	3	2	3	2	-	-	2	2	2	2	3		2	3	2
CO5	2	3	3	2	2	2	-	-	3	3	3	3		2	3	3

* 1 = Low; 2 = Moderate; 3 = Strong

Course Code	25CAP320B	L	T	P	C
Course Title	Agile Methodologies	4	-	-	4
Major Elective Course II		Employability, Entrepreneurship & Skill Development			
Pre-requisite	Introductory exposure to software development life cycle and management principles.	Regulation 2025-2026			
Class	II MCA	Semester	III		
Course Outcomes	On completion of this course, student will be able to:		K Level		
	CO1: implement Agile mindset and principles in real-world software development situations.		K1, K2, K3		
	CO2: examine and interpret the effectiveness of Scrum practices in iterative software development environments.		K1, K2, K3, K4		
	CO3: analyze and apply Extreme Programming (XP) practices to improve software development processes in dynamic requirement environments.		K1, K2, K3, K4		
	CO4: evaluate Lean and Kanban practices to optimize workflow, eliminate waste, and improve value delivery in software projects.		K1, K2, K3, K4, K5		
	CO5: design a strategic Agile transformation approach integrating leadership, coaching, and cultural dimensions of organizational change.		K1, K2, K3, K4, K5, K6		

SYLLABUS

Module	Contents	No. of Hours
I	Agile Foundations, Values and Principles: Learning Agile: Meaning and Significance of Agile in Software Development - Agile as a Mindset Rather Than a Methodology - Understanding Complexity, Change and Uncertainty. Understanding Agile Values: Perspectives from Teams and Organizations - Agile Values Explained: Individuals and Interactions - Working Software as a Measure of Progress - Customer Collaboration Over Contract Negotiation - Responding to Change Over Following a Plan - Balancing Value, Quality, and Predictability. The Twelve Agile Principles: Detailed Interpretation - Delivering Value Early and Continuously - Embracing Change for Customer Advantage - Sustainable Development and Technical Excellence - Face-to-Face Communication and Collaboration.	10
II	The Scrum Framework: Scrum and Self-Organizing Teams: Scrum Overview - Empirical Foundation - and its Emergent Design. Scrum Roles: Product Owner, Scrum Master, Development Team - Building and Managing the Product Backlog - Sprint Planning and Collective Commitment-Forecasting, Estimation and Shared Accountability.	9
III	Extreme Programming (XP): Introduction to Extreme Programming: XP as a Response to Rapidly Changing Requirements. Core XP Values: Communication, Simplicity, Feedback, Courage, Respect - Incremental Design and Continuous Improvement - Embracing Change Through Short Feedback Loops. Simplicity as a Design Strategy: Collaboration and Shared Ownership in XP Teams - Quality Built-In Versus Inspected-In - Strengths, Limitations, and Applicability of XP.	10

IV	Lean and Kanban: Origins of Lean Thinking and Its Relevance to Software: Seeing the Whole System in Software Development - Eliminating Waste and Maximizing Value - Value Stream Mapping Concepts – Gain a deeper understanding of the Product. Introduction to Kanban Method and Workflow: The Principles of Kanban - Visualizing Work and Workflow Policies - Limiting Work-in-Progress (WIP) - Managing Flow and Lead Time - Continuous Improvement Through Evolutionary Change.				10
V	Agile Coaching and Organizational Transformation: The Agile Coach: Roles, responsibilities, and competencies - Coaching mindset versus traditional management mindset - Teaching, mentoring, and facilitating in Agile environments - Building trust, collaboration, and psychological safety in teams. Leadership transformation in Agile organizations: Organizational culture and values alignment - Common patterns of failure in Agile adoption - Sustaining continuous improvement and learning cultures. Case Study: Agile Transformation in a Traditional Organization.				9
	Total Hours				48
References	Essential: [Text Book] 1. Andrew Stellman and Jennifer Greene, “Learning Agile: Understanding Scrum, XP, Lean, and Kanban” O’Reilly Media, Fifth Indian Reprint, October 2021 [Modules: I to V]. Suggestive: 1. Mike Cohn, Succeeding with Agile: Software Development Using Scrum, Foreword by Tim Lister, Pearson Education, Tenth Impression, 2018. 2. Orit Hazzan, Yeal Dubinsky, “Agile Software Engineering”, Springer International Edition, Second Indian Reprint, 2014. 3. Lyssa Adkins, “Coaching Agile Teams: A Companion for ScrumMasters, Agile Coaches, and Project Managers in Transition”, Pearson Education India, 2012. Online Resources: 1. Agile Values and Twelve Principles: https://agilemanifesto.org/ 2. Scrum framework, roles, events, and artifacts: https://scrumguides.org/ 3. XP values, practices, simplicity, feedback loops: http://www.extremeprogramming.org 4. Kanban principles, WIP limits, workflow visualization: https://kanban.university/resources/ 5. Agile estimation, planning, backlog refinement: https://www.mountangoatsoftware.com/ 6. Agile transformation and enterprise agility concepts: https://www.scaledagileframework.com				
Teaching-Learning Process	1. Chalk and Talk 2. PowerPoint Presentation 3. E-learning/E-content 4. Seminars (talks by experts) 5. Quiz				
Assessment Methods	1. Open-book tests 2. Problem based assignments 3. Case-study reports 4. Team project reports 5. Oral presentation including seminar presentation				
Designed by	Dr.L. Thara	Verified by	Dr.C. Thiyagarajan	HoD	Dr.L. Thara

	<i>PO1</i>	<i>PO2</i>	<i>PO3</i>	<i>PO4</i>	<i>PO5</i>	<i>PO6</i>	<i>PO7</i>	<i>PO8</i>	<i>PO9</i>	<i>PO10</i>	<i>PO11</i>	<i>PO12</i>		<i>PSO1</i>	<i>PSO2</i>	<i>PSO3</i>
<i>CO1</i>	3	2	3	3	2	2	-	-	2	-	2	2		3	2	-
<i>CO2</i>	3	2	3	3	3	2	3	2	-	3	-	-		3	2	3
<i>CO3</i>	3	2	3	3	3	2	3	3	-	3	1	3		3	2	3
<i>CO4</i>	3	2	3	3	3	3	-	3	2	-	3	2		3	3	-
<i>CO5</i>	3	3	3	3	3	3	2	-	3	1	3	3		3	3	2

Mapping

* 1 = Low ; 2 = Moderate ; 3 = Strong

Course Code	25CAP321	L	T	P	C
Course Title	Practical-VII (Artificial Intelligence Lab)	-	-	3	2
Major Course		Employability, Entrepreneurship & Skill Development			
Pre-requisite	Problem solving & Programming, Basics of Machine Learning	Regulation 2025-2026			
Class	II MCA	Semester	III		
Course Outcomes	On completion of this course, students will be able to			K Level	
	CO1: apply good programming methods for searching techniques			K1, K2, K3	
	CO2: evaluate the AI models pre-processed through various feature engineering algorithms by Python Programming.			K1, K2, K3, K4, K5	
	CO3: create various game-based applications to solve real-time complex problems.			K1, K2, K3, K4, K5, K6	

SYLLABUS

Contents					
LIST OF EXPERIMENTS					No. of Hours
<u>Implement the following concepts in Python to:</u> 1. Build Breadth First Search. 2. Develop Depth First Search. 3. Construct Tic-Tac-Toe game. 4. Develop 8-Puzzle problem. 5. Build Water-Jug problem. 6. Develop Travelling Salesman Problem. 7. Demonstrate Tower of Hanoi. 8. Develop Monkey Banana Problem. 9. Create Alpha-Beta Pruning. 10. Develop 8-Queens Problem. 11. Create simple Chatbot. 12. Develop Hangman game. 13. Eliminate stop words for a given passage from a text file using NLTK with Sentiment Analysis.					36 Hours
Designed by	Dr.S. Manju	Verified by	Dr.S. Anita Sofia	HoD	Dr.L. Thara

Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12		PSO1	PSO2	PSO3
CO1	3	2	3	3	3	3	3	2	3	3	2	3		3	2	3
CO2	3	3	2	3	3	3	3	2	3	3	2	3		3	3	3
CO3	3	3	2	2	2	2	2	3	3	3	2	2		3	3	2

* 1 = Low; 2 = Moderate; 3 = Strong

Course Code	25CAP322			L	T	P	C
Course Title	Practical - VIII (Machine Learning Lab) [Skill based]			-	-	2	2
Major Course				Employability & Skill Development			
Pre-requisite	Problem Solving and Programming Lab		Regulation 2025-2026				
Class	II MCA		Semester		III		
Course Outcomes	On completion of this course, students will be able to:				K level		
	CO1: explain machine learning concepts, data preprocessing techniques, and model evaluation methods.				K1, K2, K3		
	CO2: apply supervised and unsupervised machine learning algorithms to build and analyse predictive models on real datasets.				K1, K2, K3, K4, K5		
	CO3: implement and optimize machine learning models using Python libraries and tools for practical applications.				K1, K2, K3, K4, K5, K6		

SYLLABUS

Contents					
LIST OF EXPERIMENTS					No. of Hours
1. Import a dataset using Pandas to handle missing values, encode categorical variables, and perform feature scaling and normalization. (Use a dataset related to medicinal plants used in Ayurveda). 2. Perform a statistical summary to visualize data distributions using Matplotlib/Seaborn, Plot correlation heatmaps, and identify outliers 3. Split the dataset into training and testing sets and apply k-fold cross-validation to compare model performance across folds 4. Implement simple and multiple linear regression to compute MSE and R^2 score, and plot the regression line 5. Build a logistic regression classifier to perform binary classification and evaluate using a confusion matrix and accuracy. 6. Implement KNN classifier, test with different K values, and compare the accuracy results. Apply the model for classification of traditional plant species based on morphological features. 7. Compare the performance of the Decision Tree & Random Forest 8. Implement Gaussian Naïve Bayes, evaluate precision, recall, and F1 score 9. Train SVM with different kernels, compare linear vs RBF kernel performance 10. Apply K-Means clustering to determine optimal K using the elbow method, and perform hierarchical clustering and dendrogram visualization 11. Apply PCA to reduce features, compare model performance before and after PCA 12. Perform hyperparameter tuning using Grid Search, build an end-to-end ML pipeline and train, evaluate, and report results.					24
Designed by	Dr. V S Anita Sofia		Verified by	Dr. S Manju	HoD Dr. L Thara

Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12		PSO1	PSO2	PSO3
CO1	3	2	3	3	3	3	3	2	3	3	2	3		3	2	3
CO2	3	3	2	3	3	3	3	2	3	3	2	3		3	3	3
CO3	3	3	2	2	2	2	2	3	3	3	2	2		3	3	2

* 1 = Low; 2 = Moderate; 3 = Strong

Course Code	25CAP323		L	T	P	C
Course Title	Practical-IX (Full Stack Development Lab) [Skill based]		-	-	4	4
Major Course			Employability, Entrepreneurship & Skill Development			
Pre-requisite	Knowledge of C, OOPS, Java, Python, Web Framework		Regulation 2025-2026			
Class	II MCA		Semester		III	
Course Outcomes	On completion of this course, students will be able to:			K Level		
	CO1: apply the client-side of the web application using JavaScript.			K1, k2, K3		
	CO2: create an application with JavaScript on the desktop using HTML, ReactJS, and jQuery.			K1, K2, K3, K4, K5		
	CO3: create new applications by applying the drag-and-drop method.			K1, K2, K3, K4, K5K6		

SYLLABUS

Contents					
LIST OF EXPERIMENTS					No. of Hours
1. Create a simple webpage using HTML. 2. Develop a website using HTML, CSS, and JavaScript. 3. Create a form and validate the contents of the form using JavaScript. 4. Build a Chat module using HTML, CSS, and JavaScript. 5. Build a desktop application on node.js to display index.html page. 6. Create a simple calculator Application using ReactJS. 7. Implement an application to count visitors of the website using session 8. Develop a voting application using React JS 9. Build a Password Strength Check using jQuery 10. Implement a star rating system using jQuery 11. Create a Simple Login form using ReactJS and deployment using Docker container. 12. Build a program to validate the following fields in a registration page i. Name (start with alphabet and followed by alphanumeric, and the length should not be less than 6 characters) ii. Mobile (only numbers and length 10 digits) iii. E-mail (should contain format like xxxxxxxx@xxxxxxx.xxx) 13. Using the CMS, users must be able to design a web page using the drag-and-drop method 14. Create a project on a Grocery delivery application, managing using Kubernetes. 15. Create an e-commerce project portal for used furniture sales. Assume this project is for a startup that acquires used furniture from users at a price, refurbishes it, and sells it at a margin. The website must display second-hand furniture currently for sale.					48
Designed by	Dr. V S Anita Sofia	Verified by	Dr. S Manju	HoD	Dr. L Thara

Mapping

	<i>PO1</i>	<i>PO2</i>	<i>PO3</i>	<i>PO4</i>	<i>PO5</i>	<i>PO6</i>	<i>PO7</i>	<i>PO8</i>	<i>PO9</i>	<i>PO10</i>	<i>PO11</i>	<i>PO12</i>		<i>PSO1</i>	<i>PSO2</i>	<i>PSO3</i>
<i>CO1</i>	3	2	3	3	3	3	3	2	3	3	2	3		3	2	3
<i>CO2</i>	3	3	2	3	3	3	3	2	3	3	2	3		3	3	3
<i>CO3</i>	3	3	2	2	2	2	2	3	3	3	2	2		3	3	2

* *1 = Low; 2 = Moderate; 3 = Strong*

Course Code	25IDCCAPT (IDC for Cluster PG)	L	T	P	C
Course Title	Data Analytics	3	-	-	2
Interdisciplinary Course – Theory		Employability & Skill Development			
Pre-requisite	Basic programming (Python/R), basic probability, and introductory databases.	Regulation 2025-2026			
Class	II MCA	Semester	III		
Course Outcomes	On completion of this course, student will be able to:			K Level	
	CO1: define analytics problems and select suitable data sources with ethical constraints.			K1, K2, K3	
	CO2: clean, transform, and engineer features using reproducible preprocessing workflows.			K1, K2, K3	
	CO3: perform EDA and apply core statistical reasoning to justify insights and decisions.			K1, K2, K3, K4	
	CO4: build supervised and unsupervised models and validate them using proper evaluation protocols.			K1, K2, K3, K4, K5	
	CO5: interpret results, communicate findings effectively, and propose responsible, actionable recommendations.			K1, K2, K3, K4, K5	

SYLLABUS

Module	Contents	No. of Hours
I	Data Analytics Foundations and Data Understanding: Data analytics vs data science - Types of analytics: descriptive, diagnostic, predictive, prescriptive - Analytics lifecycle: problem framing, data, model, deployment and monitoring - Data types and measurement: Structured, semi-structured, unstructured data - Sampling basics and data quality dimensions - Data sources: files, APIs, databases, logs, sensors - Data ethics: consent, privacy, bias, fairness, transparency - Reproducibility and versioning with datasets	7
II	Data Preprocessing: Data cleaning: Missing values - MCAR/MAR/MNAR intuition - imputation strategies - Outliers: IQR, Duplicates, inconsistent formats, noisy labels - Data transformation: Encoding - one-hot, ordinal, target (with leakage warnings) - Scaling - standardization, min-max, robust scaling.	7
III	Feature Engineering: Feature construction from time, text (basic), categories - Feature selection: filter, wrapper, embedded - Data leakage, train/validation split discipline, pipelines.	7
IV	Exploratory Data Analysis and Statistical Inference: EDA workflow- Summary statistics, distribution analysis, correlation and dependence - Visualization choices: histograms, boxplots, scatter, heatmaps - Statistical inference essentials: Confidence intervals and hypothesis testing (t-test, chi-square intuition)	7
V	Analytics communication: Storytelling with data: audience, narrative, chart integrity - Dashboard principles (KPIs, filters, drill-down) - Reporting: OMRL Structure – Data Analytics best practices (chart selection and readability) – Designing simple analytical reports and presentation of insights – Communicating analytical findings to technical and non-technical audiences.	8
	Total Hours	36

References	Essential: [Text Book] 1. T. A. Runkler, Data Analytics: Models and Algorithms for Intelligent Data Analysis – A Comprehensive Introduction, Latest edition (2025), Springer. [Modules: I, II] 2. J. Han, M. Kamber, J. Pei, <i>Data Mining: Concepts and Techniques</i>, 4th edition (2022), Morgan Kaufmann/Elsevier. [Modules: III, IV] 3. Wes McKinney, Python for Data Analysis: Data Wrangling with pandas, NumPy, and Jupyter, 3rd ed. (2022), O'Reilly. [Module V] Suggestive: 1. G. James, D. Witten, T. Hastie, R. Tibshirani, J. Taylor, An Introduction to Statistical Learning, (2023 release listed), Springer. 2. Anil Maheshwari, <i>Data Analytics</i>, 2nd edition (2023), McGraw Hill Education (India). 3. P. Chavan, K. Pampattiwar, R. Mangrulkar, Essentials of Big Data Analytics: Applications in R and Python, 1st edition (Jan 2026), Morgan Kaufmann/Elsevier. Online Resources: 1. https://epgp.inflibnet.ac.in/ 2. https://onlinecourses.nptel.ac.in/noc25_cs17/preview 3. https://www.geeksforgeeks.org/data-analysis/data-analysis-tutorial/	
Teaching-Learning Process	1. Chalk and Talk 2. Power Point Presentation 3. E-Learning / E-Content 4. Real-Time Workshops 5. Audio and Video Tools	
Assessment Methods	1. Seminar Presentation 2. Problem Based Assignments 3. Case Presentations & Demonstrations 4. Open Books Test 5. Team Projects	
Designed by	Dr.S.Nithyanandh	Verified by Dr.C.Thiyagarajan HoD Dr.L.Thara

Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12		PSO1	PSO2	PSO3
CO1	3	3	3	3	2	3	3	2	3	3	3	3		3	2	3
CO2	3	2	3	3	3	3	3	3	2	3	2	2		3	2	3
CO3	3	2	3	3	3	3	2	3	2	3	2	3		3	3	2
CO4	3	2	3	3	3	3	2	3	2	2	2	2		3	2	3
CO5	3	3	3	3	3	2	3	3	3	2	3	3		3	3	3

* 1 = Low ; 2 = Moderate ; 3 = Strong

Course Code	25IDCCAPP (IDC for Cluster PG)	L	T	P	C
Course Title	Data Analytics Lab [Skill Based]	-	-	2	2
Interdisciplinary Course – Practical		Employability & Skill Development			
Pre-requisite	Basic Programming Knowledge in Python, Database Concepts and Fundamental Knowledge of Statistics	Regulation 2025-2026			
Class	II MCA	Semester	III		
Course Outcomes	On completion of this course, students will be able to			K Level	
	CO1: apply appropriate data preprocessing techniques such as handling missing values, noise removal, duplicate elimination, and imputation to prepare datasets for analysis.			K1, K2, K3	
	CO2: build and evaluate basic machine learning models including Linear Regression, Logistic Regression, Decision Tree, and Random Forest using suitable performance metrics.			K1, K2, K3, K4	
	CO3: interpret analytical results using visualization and descriptive statistical measures to derive meaningful insights from real-world datasets.			K1, K2, K3, K4	

SYLLABUS

Contents	
LIST OF EXPERIMENTS	No. of Hours
1. Create a sample dataset and generate a data dictionary (column name, data type, and description) along with basic dataset profiling (shape, column types, and missing value count). 2. Build a program to read sample data from two sources (CSV and API-like JSON sample), merge them into a single dataset, and verify data quality dimensions such as completeness and consistency. 3. Design a program to identify missing values in a sample dataset and handle them using suitable techniques (mean/median/mode filling and row removal). 4. Perform feature engineering on sample data by creating new features (e.g., extracting month from date, converting category to one-hot encoding) and applying scaling (min–max or standardization). 5. Implement Exploratory Data Analysis (EDA) on a sample dataset by generating summary statistics and plotting histogram and boxplot for numerical columns. 6. Develop a Program to perform correlation analysis (t-test) on a sample dataset and interpret the result using a short conclusion statement. 7. Build a Program to perform simple hypothesis testing (chi-square) on a sample dataset and interpret the result using a short conclusion statement. 8. Develop a program using Google Colab or an alternative tool to create a Line Chart representing monthly sales performance using an E-Commerce dataset. 9. Create a Bar Chart that visualizes the number of students enrolled in different departments using a sample dataset. 10. Create a simple KPI dashboard table (Total, Average, Growth %, Best Month) from sample sales data and present insights in the OMRL reporting format (Objective–Method–Result–Limitation).	24

11. Perform data visualization techniques on sample data using Bar chart, Line chart, Scatter plot, and Pie chart. 12. Implement descriptive analytics on a sample healthcare dataset (mean, median, mode, min, max, standard deviation, and basic insights).					
Designed by	Dr.S. Nithyanandh	Verified by	Dr.C. Thiyagarajan	HoD	Dr.L. Thara

Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12		PSO1	PSO2	PSO3
CO1	3	3	2	3	3	2	2	3	2	2	3	2		3	2	3
CO2	3	3	3	3	3	3	2	3	2	2	2	2		3	3	3
CO3	3	3	3	2	3	2	2	2	3	2	3	2		3	3	2

* 1 = Low; 2 = Moderate; 3 = Strong